Zero-Shot and Fine-Tuned NER in European Portuguese Chest CT Reports

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Abstract

Radiology reports in Portugal are written in free text, and Named Entity Recognition (NER) is the first step towards structuring the relevant information they contain. The corpora and biomedical NER models available for Portuguese are, in their majority, built on the Brazilian variety of the language. The only comparison we are aware of on European Portuguese (EP) radiology reports covers those models alone and leaves their behaviour before any task-specific training unmeasured. We evaluate 8 encoder-only models on 68 chest Computed Tomography (CT) reports containing 324 entities, using zero-shot and fine-tuning strategies, with GLiNER-BioMed as an additional zero-shot baseline. Under relaxed matching, the fine-tuned models reach F_1 between 0.797 and 0.875, whereas the best untrained model reaches 0.625. Models that arrive with a fixed label set inherited from another corpus predict almost nothing before training, yet become competitive once the classification head is replaced, showing that the limitation lies there and not in the underlying representations. Zero-shot performance is therefore a poor predictor of the ranking obtained after fine-tuning. Continued pre-training on medical text yields only 0.015 more in F_1 than the same model without it, a difference within the spread across folds. Annotated data matters more here than the choice of model.

1 Introduction

Radiology reports in Portugal are typically written in free text, which leaves the content unavailable to any system that is not prepared for unstructured data. Structuring it would let a service count, retrieve and compare findings across exams without a radiologist reading each report again to track over successive studies [1]. NER is the first step towards structuring that content, marking the segments of text that designate clinically relevant concepts [2]. In chest CT, those concepts are the findings, the anatomical structures involved and the medical devices visible in the image, such as pacemakers, stents and catheters, which must be identified before reports can serve cohort selection or registries. To recognize entities, annotated corpora and pre-trained models are required, and EP is short of both, since resources labelled as Portuguese are dominated by the Brazilian variety [3, 4], with the first medical language model for the European variety appearing only in 2024 [5]. Existing benchmarks for clinical NER in Portuguese are, in their majority, on Brazilian clinical notes [6], and clinical NER in EP has been addressed in other medical fields [7, 8]. To our knowledge, none of these measures how the models behave before task-specific training.

When annotating a new corpus, selecting a pre-trained model for fine-tuning based on the performance of untrained candidates is only useful if zero-shot performance predicts performance after fine-tuning. We therefore fine-tuned eight models on 68 annotated EP chest CT reports and evaluated five zero-shot setups: four models with pre-existing clinical NER heads and GLiNER-BioMed [9], prompted with the target entity types. The 4 models with pre-existing heads were evaluated both off-the-shelf and after fine-tuning. Both regimes use 5-fold cross-validation, split by report, so no report contributes to training and testing at the same time. The two regimes rank the models differently, and the F_1 gap between regimes exceeds the spread among trained models.

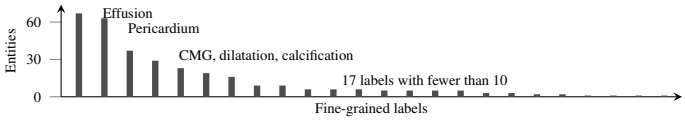


Figure 1: The 24 fine-grained labels, ranked by frequency.

2 Methods

Corpus. A radiology expert annotated 68 EP chest CT reports. The reports contain 2036 sentences and 31205 tokens (15.3 tokens per sentence on average, SD 16.2, median 9), of which only 160 sentences carry annotations, yielding 324 entities. There are three entity types: FINDING, STRUCTURE and DEVICE, with 186, 129 and 9 entities respectively, with the addition of a binary feature recording whether a finding is asserted or negated. Figure 1 shows the frequency of the 24 fine-grained labels. The distribution is heavily skewed: 17 labels have fewer than 10 examples and 4 have a single one, which makes training at that level of granularity impractical.

Models. In total, 8 models were fine-tuned, and all are encoder-only models. MediAlbertina PT-PT 900M [5] is the first publicly available model adapted to the biomedical setting in EP, and albertina-900m-ptpt [10] is its counterpart without any medical pre-training. BioBERTpt-all and the clinicalnerpt family (110M each) were trained on Brazilian Portuguese [4]. EuroBERT-210m [11] is a general multilingual base model and ModernBERT-base (150M) [12] an English one, both without clinical specialisation. All models share the same training budget of 10 epochs at a learning rate of 3×10^{-5}. MediAlbertina is the single exception, because at that rate it collapsed in 3 of its 5 folds and predicted one label throughout, so it was given a lower rate of 1×10^{-5} over 15 epochs. The code is available online.[1] The reports cannot be released due to patient privacy.

Evaluation. We report precision, recall and F_β for $\beta \in \{0.5, 1, 2\}$ [6]. We report three values of β because missing a finding is costlier than proposing one for review, but spurious spans are not harmless either. Defining whether a predicted entity counts as correct led us to use two matching criteria: a strict match requiring exact boundaries and the correct label over the whole report, and a relaxed match (restricted to annotated sentences) that disregards labels and accepts any shared token with a reference span. Figure 2 illustrates the two on a sentence from the corpus. Strict match is the more precise of the two, so it carries our main results. Relaxed match exists because the untrained models cannot be scored under the strict criterion at all, as they emit the labels of the corpora they were trained on rather than FINDING, STRUCTURE and DEVICE. Comparing the two regimes requires a criterion that both can satisfy, and this is what relaxed match provides. The zero-shot experiment therefore becomes a span detection task rather than NER, since it does not ask whether a label is predicted correctly, but whether these models locate the right spans of text at all. Models whose label sets differ from ours were used deliberately, since no EP radiology NER model exists, and a disease or a sign in a clinical corpus corresponds to a finding here. The zero-shot regime therefore reflects a team that annotates a small evaluation set to screen models before investing in training data.

[1] https://github.com/LIAAD/NER_RADIOLOGIA_PT

We used 5-fold cross-validation with a fixed seed, testing each report exactly once. In each fold, 40 to 42 reports were used for training, 13 to 14 for validation (monitoring loss), and 13 to 14 for testing. On an NVIDIA A100 GPU, 5-fold training took 8 to 18 minutes for most models and 75 minutes for MediAlbertina (15 epochs).

3 Results and Discussion

Zero-shot. Under relaxed match, the only criterion the two regimes share, the models fine-tuned on our annotations obtain F_1 scores ranging from 0.797 to 0.875, whereas the best result obtained without any training is 0.625. GLiNER-BioMed, the only model here built for zero-shot use, reaches 0.589. Relaxed match rewards locating the span, not naming it, which is how an untrained model scores above zero. Even the worst fine-tuned model, ModernBERT-base at 0.797, remains 0.172 above any model in its untrained form. Two of the 3 clinicalnerpt variants have negligible results before training (correctly identifying 5 and 2 of the 324 entities in clinicalnerpt-finding and clinicalnerpt-sign respectively). Even so, all 3 models become competitive after training, with clinicalnerpt-finding rising from 0.026 to 0.872. If the representations were unsuited to EP radiology, replacing the classification head would not lift a model from near zero to the top of the table, so the limitation lies in the label set inherited from the models' origin. Zero-shot results are therefore not a reliable predictor of how a model will perform after fine-tuning in this domain.

Fine-tuning. Table 1 reports the models under the strict criterion. Under F_1, MediAlbertina leads (0.726), although its margin over albertina-900m-ptpt is small (0.015) and may partly reflect a lower learning rate (single seed). The ordering is not stable across the three F_β scores, since under $F_{0.5}$ EuroBERT-210m takes the lead at 0.741, combining the best precision of the set with a low recall. All models but ModernBERT-base fall within 0.035, an interval comparable to the spread across folds, so their ordering can be suggested but not asserted. Training clinicalnerpt-finding on 10, 20, 31, and 41 reports yields strict F_1 of 0.566, 0.674, 0.682, and 0.714. The non-flattened curve suggests annotations remain the bottleneck.

Table 1: Fine-tuned results, strict criterion, averaged over 5 folds. SD is the standard deviation of F_1 across folds. Bold marks the best value in each metric column.

Model	P	R	$F_{0.5}$	F_1	F_2	SD
MediAlbertina	0.724	**0.730**	0.725	**0.726**	**0.729**	0.024
clinicalnerpt-finding	0.725	0.709	0.721	0.715	0.711	0.030
albertina-900m-ptpt	0.738	0.688	0.727	0.711	0.697	0.036
clinicalnerpt-disease	0.739	0.688	0.727	0.711	0.697	0.037
clinicalnerpt-sign	0.709	0.705	0.707	0.704	0.704	0.043
EuroBERT-210m	**0.770**	0.650	**0.741**	0.703	0.670	0.039
biobertpt-all	0.698	0.689	0.695	0.691	0.689	0.051
ModernBERT-base	0.768	0.604	0.726	0.674	0.630	0.008

Limitations and future work. The corpus is small (68 reports by a single expert, so no agreement metrics) and the labels are imbalanced. Recall is low on the rarest labels (for example zero on sternal wires and pacemakers), and the 17 labels with fewer than 10 examples hold 70 of the 324 entities, which prevents reliable per-class reporting. Future directions are the assertion versus negation distinction, relation extraction between findings and dedicated EP radiology foundation models.

4 Conclusion

The most relevant result is that even the worst fine-tuned model achieves higher performance on the studied EP radiology reports than the best untrained model, since the difference between the two is 0.172. It is important to note that medical adaptation, in this case, did not yield results much superior to the general-purpose version, since MediAlbertina is only 0.015 above the same model without medical pre-training, and is surpassed by it in precision. This is consistent with Elvas et al. [7], where albertina-900m-ptpt also edges out MediAlbertina on EP cardiology reports. Future research on clinical NER in EP should therefore allocate effort to annotation rather than to model selection.

	Aorta	torácica	de	normal	calibre	e	com	trajecto	regular	.
Gold	STRUCT	STRUCT		FIND	FIND					
Pred.	STRUCT	STRUCT			FIND					

Figure 2: Gold spans and predictions of the best model. The sentence translates as *Thoracic aorta of normal calibre and regular course*.

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